Nguyễn Văn Tiệm

Embedded Software Engineer

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Professional Summary

> **5+ years** of embedded system software development and IoT with US and European companies · Experienced in risk management, code review, and teamwork across multi-functional teams.

Technical Expertise

Category	Details
Languages	C, C++, Python, Java, Kotlin, Bash, Makefile
Microcontrollers	ESP32, STM32, Nordic nRF52, Silabs, Renesas, Toshiba, Abov, AVR
Wireless Tecnologies	Bluetooth, OpenThread, Zigbee, Z-Wave, Sidewalk, Wi-Fi , Cellular
Protocol	MQTT, WebSocket, HTTP, BLE, Matter over Thread, 2G, 4G follow 3GPP standard
Peripherals	UART, I2C, SPI, ADC, DAC, DMA, Timer, PWM
ARM cortex	Systick, NVIC, Trustzone, MPU
Embedded OS	ZephyrOS, FreeRTOS
Coding Standard	MISRA C
IDE	Keil, IAR, Simplicity Studio 5, STM32CubeIDE, E2 Studio, Arduino IDE, Eclipse
Toolchain/Compiler	GCC ARM, ESP-IDF, nRF Connect SDK
Build Tool	Make, Ninja
Debuger	IAR, STM32IDE, Ozone Debugger, Logic Analyzer
Version Control System	Git (GitHub/GitLab), SVN, Jira, Docker, Linux, , Logic Analyzer
Algorithms & DS	BFS, DFS, Backtracking, Heap, Stack, Queue, Tree, Trie, Priority Queue, Hash Map
Languages	English (Reading technical docs, basic communication)

Professional Experience

FPT Software — Embedded Developer

`04/2026 – present`

Project: Developed firmware for a cellular modem management module in a smart meter, supporting both 2G and 4G network technologies.

- Implemented the modem FOTA (Firmware Over-The-Air) feature.
- Implemented UART DMA using the linked-list DMA mode on the STM32H573.
- Implemented the modem power control flow.
- Produced software design documentation based on system requirements, including software architecture, data flow diagrams, sequence diagrams, state machines, and detailed flowcharts for each state.
- Proposed technical and algorithmic solutions and provided implementation guidance to team members.
- Reviewed code written by team members and provided technical feedback.
- Optimized RAM usage, including static RAM and stack memory, as well as Flash memory usage.

MCU: STM32H573
OS: FreeRTOS
Document Management System: Confluence
VCS: GitLab, Jira
Language: C, C++, Python
IDE/Debuger: Keil-C

LGEDV — Main Embedded Developer

`04/2025 – 04-2026`

Project: Washing machine firmware for wash-only, wash towel, and wash & dry (Heat pump) models.

- Analyzed legacy code flows and designed new logic ensuring backward compatibility
- Analyzed Black Box, verify that the product met all functional requirements and operated in accordance with the Software Requirements Specification (SRS).
- Implemented a PWM timer-based audio playback module to generate notification melodies according to the Product Requirements Document (PRD), and developed drum light effects with configurable brightness levels.
- Developed an anti-vibration control algorithm for a washing machine during high-speed spin cycles using the BMI320 inertial sensor to detect and mitigate excessive vibration.
- Stored user configuration data in EEPROM; synchronized machine status to server
- Wrote and executed test cases before handover to SQE team
- Fixed bugs in legacy firmware versions; reviewed code for team members

MCU: Abov, Toshiba, Renesas MCUs
Document Management System: colab
VCS: SVN, GitLab, Jira
Language: C, Python
IDE/Debuger: IAR, E2 Studio

FPT Software — Embedded Developer

`12/2023 – 04/2025`

Project: Garage door with BLE, Wi-Fi & Matter wireless integration (ZephyrOS, nRF platform).

- Defined device tree for custom board; performed board bring-up from schematics
- Implemented OTA via BLE with custom external flash; OTA for motherboard via BLE & UART
- Implemented ECDH security to establish secure BLE channel; designed BLE message protocol
- Integrated FAT & LittleFS for user/door config/log management
- Ported Matter Door Lock & Window Covering profiles; launched Matter Hub on Raspberry Pi 4
- Developed the RX8130CE RTC (I^2C) driver and implemented a time-based scheduling engine to execute scheduled actions, such as automatic door opening and closing at predefined times.
- Integrated the NINA-W132 Wi-Fi module and implemented MQTT communication with AWS IoT Core for secure remote device monitoring and control.
- Optimized RAM/Flash usage; tuned mutex, queue, and buffer performance

Tech: C, C++, Python, Bash, Makefile · ZephyrOS · BLE, Matter, Wi-Fi · Docker, RPi, AWS IoT Core · Jira, Git, nRF Connect, Ozone debugger

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FPT Software — Embedded Engineer (R&D)

`10/2023 – 11/2023`

Focus: Nordic SDK, Zigbee, Sidewalk, Matter over Thread

- Studied Nordic SDK; implemented Zigbee switch/light cluster on dev board
- Prototyped Matter Door Lock on Silicon Labs; launched Matter Hub on RPi & laptop
- Built bootloader and launched Linux on BeagleBone
- Upgraded SDK and fixed issues in existing door lock project

Tech: C, C++ · Zigbee, Sidewalk, Matter/Thread · Simplicity Studio, Docker, GitHub

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FPT Software — Embedded Engineer

`04/2023 – 09/2023`

Project: Door Lock with Z-Wave wireless technology.

- Analyzed and optimized device power consumption
- Mapped Z-Wave command classes to TLV format for customer integration
- Set up Z-sniffer and self-test tools for Z-Wave certification process

Tech: C, Python · Z-Wave, RTOS · Simplicity Studio, Ozone, Logic Analyzer

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Samsung Vietnam Mobile Centre — Android Developer

`04/2021 – 03/2023`

Projects: Samsung app maintenance, PDF reader app, automated testing app.

- Maintained Samsung production applications
- Developed a full-featured PDF reader application
- Built an automated testing Android application

Tech: Java, Kotlin · Android SDK (Activity, Fragment, Service, BroadcastReceiver) · RxJava, SQLite, Design Patterns

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Personal Projects

Project	Description
Secure Bootloader & DFU — STM32	Designed and implemented a secure bootloader for the STM32F103 using ECDSA with the NIST P-256 curve for firmware signing and signature verification, a dual-bank firmware update mechanism, and power-loss protection to ensure secure and reliable firmware updates.
Cryto library	Designed and implemented optimized cryptographic libraries for the STM32F103, including ECDSA over NIST P-256, X25519-based ECDH key agreement, and AES-128-GCM authenticated encryption, with a strong focus on execution speed, Flash footprint, and memory efficiency.
Apply Edge AI on ESP32	Developed an embedded TinyML application on the ESP32 using Edge Impulse to train, deploy, and run a machine learning model locally for motion classification based on MPU6050 IMU sensor data. Collected and labeled motion datasets, trained and optimized the model, and implemented real-time inference to classify left, right, up, down, and circular gestures.
Keypad Driver — STM32U585	IQS7222B touch scanning + TCA8418 button LEDs via I2C
Sensor Drivers via I2C	MPU6050, LIS2, HTS221, LPS22; DMA peripheral communication on STM32F411
OTA via BLE — nRF52840	Bluetooth OTA using internal and external flash
LCD G9A01 Driver	SPI driver implementation for STM32H532
NOR Flash Winbond 25Q64	External flash driver via SPI for STM32H532

Education

B.Eng. Computer Engineering — 2016–2021
Hanoi University of Science and Technology (HUST), Vietnam

Certifications

- **STP Level Pro** — Samsung Talent Programmer
- **SPCT** — LGEDV